Rational three-torsion orbit screen

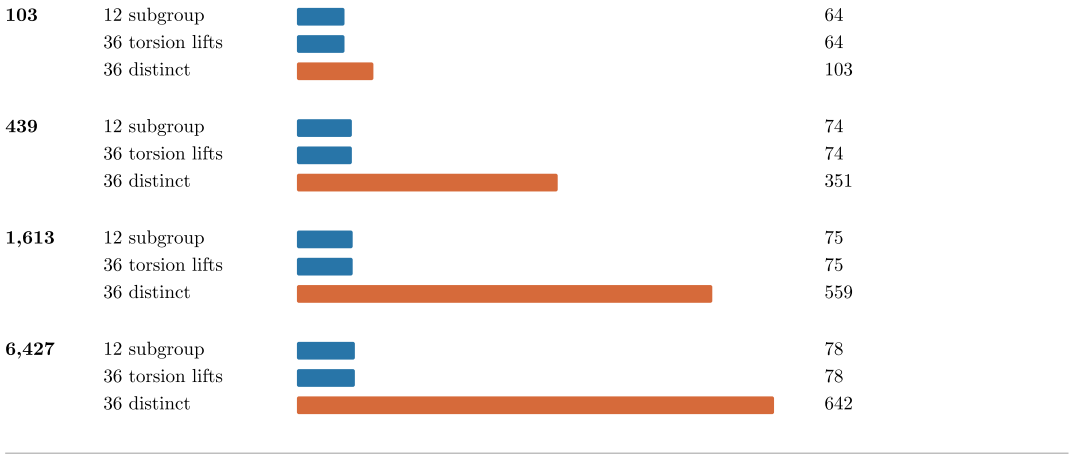
2026-10-13 local / 2026-10-14 UTC. Conductor T-2611. Source-linked full report: `REPORT.md`; frozen protocol: `PROTOCOL.md`; native counts: `raw-v1.tsv`; code: `screen.rs`; proof map: `PROOF_MAP.md`.

The tested question was whether three rational lifts of each of twelve factor-base points add new two-sum relation targets after torsion correction and projection to the prime-order subgroup. They did not: support was identical to the twelve-point control in all four groups.

Subgroup r	12 points	36 lifts	36 distinct	Lifts / 12
103	64	64	103	1.000
439	74	74	351	1.000
1,613	75	75	559	1.000
6,427	78	78	642	1.000

Exact two-sum subgroup support

Four rational three-torsion, non-isomorphic degree-three pairs. Bar length is linear in the number of distinct reachable subgroup elements; full scale 642 elements = 3.5 inches. Values come directly from `raw-v1.tsv`.



■ orbit-equivalent support; ■ independent subgroup points. Each orbit arm has 36 rational representatives but only twelve quotient classes. Counts are exact on these finite groups; this graph is a relation-support diagnostic, not whole-method speed.

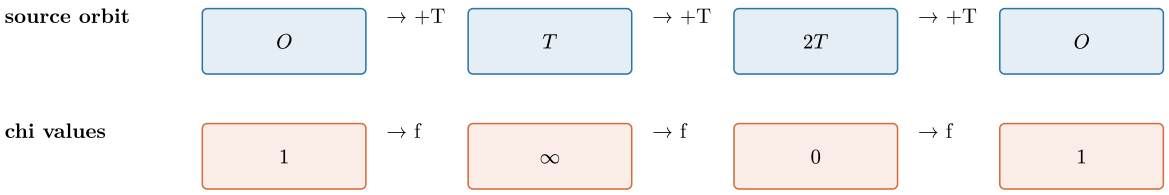
All values are exact distinct subgroup support counts. `screen.rs` checked torsion order, point distinctness, source and destination equations, isogeny and cofactor invariance, subgroup order, and point-addition against a complete scalar index. The conditional quotient-support implication built in Lean 4.34.1 with no axioms.

A nonfactorable equivariant coordinate

On `E:y²=x³+1`, let `T=(0,1)` and `chi=(y+1)/(y−1)`. The proposed action is `chi(P+T)=f(chi(P))` with `f(z)=1/(1−z)` on the projective line. The three kernel points map to `1, infinity, 0`, so this coordinate does not factor through the degree-three quotient. The action has order three.

A cubic coordinate that distinguishes the three kernel points

For each `p` in 617, 2633, 9677, 38561: source `E/F_p:y²=x³+1`, destination `E'/F_p:y²=x³−27`, `T=(0,1)`, and the verified degree-three quotient `phi:E→E'`.



`chi=(y+1)/(y−1)` and `f(z)=1/(1−z)`; the three distinct values show that `chi` does not factor through `phi`. In contrast, `phi(O)=phi(T)=phi(2T)=O` on the destination. The finite verifier checked `chi(P+T)=f(chi(P))` on all 51,492 rational source points across the four fields and checked `f³=id` on all 51,492 projective-line points.

The native Rust verifier checked both the `+T` and `+2T` identities on all 51,492 rational source points across the four fields, plus `f³=id` on all 51,492 projective-line points. Its inputs and counts are in `EQUIVARIANT_PROTOCOL.md` and `equivariant-v1.tsv`. This is a coordinate identity result; a complete extension-field oracle comparison is pending.

For generic `P`, the addition slope `(y−1)/x` gives `y(P+T)=y−2−(y−1)³/x³=(y−3)/(y+1)` on `y²=x³+1`. Conjugating this rational map by `chi` gives `f`. The two denominator-cleared polynomial steps have native identity certificates `IDC1h41adc1f4e1c5522c` and `IDC1ha0c3d83775e32754` beside the run. The points `O,T,2T` use direct projective checks. The Lean proof map contains a compiled conditional orbit-support theorem and the order-three linear-action consequence; its curve-level bridge is still open.

The earlier four-size matched rho and full cold CPU rows are in `../full_cost_20261013/REPORT.md`. This run made no new complete-solve measurement, so it has no speed or `S` ratio. The next experiment is the same-target extension-field torsion-symmetric decomposition oracle in `NEXT_GATE.md`, compared with direct source, source-orbit and invariant quotient controls. It must price construction, relation collection, linear algebra, recovery, verification and matched rho on the same target.